GLOBAL LANDSLIDE ANALYSIS

1.Project Overview

The project will involve cleaning and preparing the dataset in Excel, performing descriptive analysis, and developing interactive Power BI dashboards to visualize trends and compare landslide events across countries, time periods, triggers, and severity levels.

Project Objective

The Objective this project is to analyze Global landslide occurs across the countries and to avoid causalities and injuries in future

2.Data Source

- Source Description and Timeline: National Aeronautics and Space Administration / DATA.GOV and 1988 – 2017
- Domain: Environmental

3.Problem Statement

- To Identify which countries, have the highest number of recorded landslide events
- To analyze which months have the most recorded landslides
- Analyze the most frequently reported landslide triggers
- To analyze which landslide categories and sizes are associated with the highest reported fatalities
- To understand which geographical areas should receive greater attention in landslide and pre-planning
- To identify forecast events in near future

4. Attribute (Column /Features) Details

Attribute Name	Data Type	Description
Event Id	Integer (Number)	Unique identifier of each event
Event Date	Short Date	Occurrence date of the landslide
Event Time	Time	Occurrence time of the landslide
Event Title	String (Text)	Title of the event
Location Description	String (Text)	Landslide happened location details
Location Accuracy	Number and String	Total affected distance of the event

Landslide Category	String (Text)	It contains various categories of the landslide event
Landslide Trigger	String (Text)	It shows the reason behind the landslide
Landslide Size	String (Text)	It shows impact size of the event
Fatality Count	Integer (Number)	Death counts of the event
Injury Count	Integer (Number)	Injury count of the landslide
Country Name	String (Text)	Affected countries name
Country Code	String (Text)	Affected countries country code
Admin Division Name	String (Text)	It contains name of the affected admin divisions
Admin Division Population	Integer (Number)	It contains overall population of the affected admin division
Gazeteer Closest Point	String (Text)	Closest point from the landslide
Gazeteer Distance	Decimal Number	It shows distance between event location and closest point
Submitted Date	Short Date	Event reports submitted date
Longitude	Decimal Number	Longitude of the event location
Latitude	Decimal Number	Latitude of the event location

5. Tools & Technologies

- **Excel:** Data cleaning, transformation, and Pivot Tables.
- **Power BI:** Data modelling, DAX calculations, visualization, and interactive dashboard creation.

6. Data Pre-Processing (Excel / Power Query)

- Removed unwanted columns for analysis purpose
- Format Table using format as table

- Changed **Data types** Respectively for each column
- Used **Removing Duplicates** in event Id column – No duplicates found in my data
- Splits Event Date and Submitted Date columns used **Text to Columns by delimiter** to split date and time
- **IF(ISBLANK(F2),"Unknown",F2)** used for all text columns for handling missing values
- **PROPER(TRIM(CLEAN(N2)))** used for all text columns
- **IF(ISBLANK(K2),AVERAGE($K$2:$K$11034),K2)** used for all numerical columns to handle missing values
- **IF(OR(ISBLANK([@[Admin Division Population]]),[@[Admin Division Population]]=0),AVERAGE($O$2:$O$11034),[@[Admin Division Population]])** – formula used for handling missing population value in admin division population column
- **IF(ISBLANK([@[Submitted Date]]),[@[Event Date]],[@[Submitted Date]])** – used for handling missing values in submitted date column
- Used **VLOOKUP** formula for creating two new spreadsheets named **Impact and Location** for better analysis
- **VLOOKUP($A2,'Global Landslide'!$A:$T,2,FALSE)**
- Did **Descriptive Statistics** analysis using toolpak

Pivot Table Screenshot

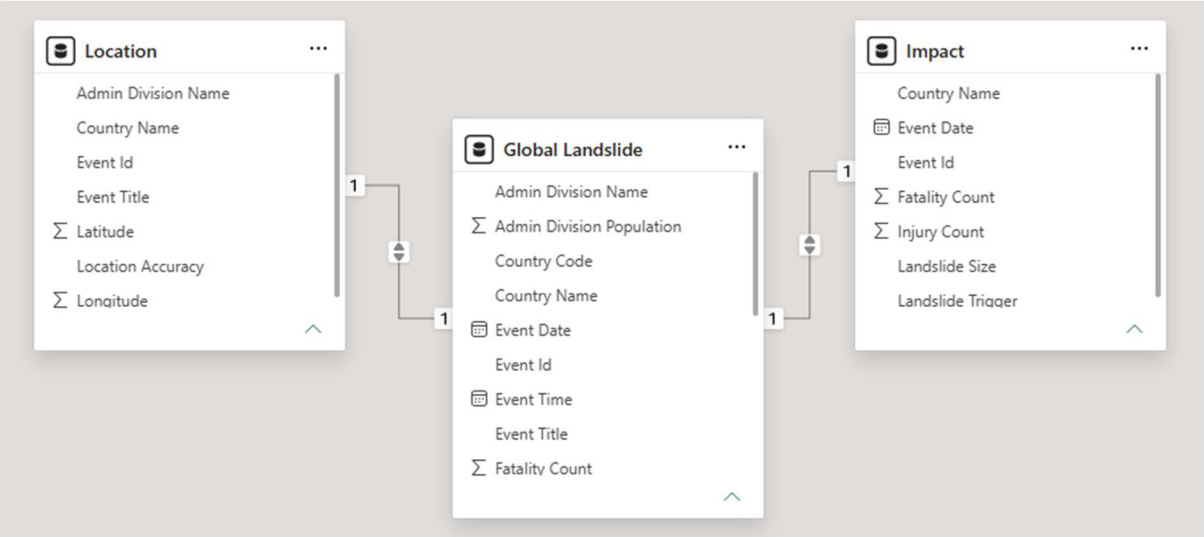
Landslide recorded in each country	
Country Name	Count of Event Id
Afghanistan	15
Albania	1
Algeria	1
American Samoa	5
Angola	3
Argentina	7
Armenia	4
Australia	105
Austria	10
Azerbaijan	21
Bangladesh	58
Barbados	1
Belize	1
Bermuda	2
Bhutan	20
Bolivia	11
Bosnia And Herzegovina	12
Brazil	214
Brunei	14

Landslide Triggers vs Fatality Count vs Injury Count		
Landslide Triggers	Sum of Fatality Count	Sum of Injury Count
Construction	149	60
Continuous_Rain	3708	428
Dam_Embankment_Collapse	312	91
Downpour	18967	4652
Earthquake	459	84
Flooding	169	9
Freeze_Thaw	38	18
Leaking_Pipe	0	0
Mining	400	95
Monsoon	774	284
No_Apparent_Trigger	15	12
Other	17	17
Rain	5418	1687
Snowfall_Snowmelt	512	91
Tropical_Cyclone	3149	443
Unknown	1034	584
Vibration	0	0
Volcano	0	0
Grand Total	**35119**	**8553**

Category vs Fatality	
Landslide Category	Sum of Fatality Count
Complex	2342
Creep	0
Debris_Flow	5821
Earth_Flow	5
Lahar	19
Landslide	19952
Mudslide	6306
Other	60
Riverbank_Collapse	18
Rock_Fall	380
Snow_Avalanche	112
Topple	0
Translational_Slide	27
Unknown	76
Grand Total	**35119**

7. Data Modelling and DAX (Power BI)

Data Model:



Calculated Column & DAX Measures:

- Total Fatality Count = sum('Global Landslide'[Fatality Count])
- Total Injury Count = sum('Global Landslide'[Injury Count])
- Total Countries Affected = DISTINCTCOUNT('Global Landslide'[Country Name])
- Total Landslides Occurred= DISTINCTCOUNT('Global Landslide'[Event Id])
- Total Affected People=[Fatality Count]+ [Injury Count]
- Average Fatalities per Event=DIVIDE(Total Fatality Count, Total Events Occurred)
- Average Injuries per Event= DIVIDE(Total Injury Count, Total Events Occurred)
- Calculated column created named "Human Impact" created in Impact Table
- Human Impact=Impact[Fatality Count]+Impact[Injury Count]

8. Analysis and Visualizations

Dashboard Features:

Visualizations based on problem statement

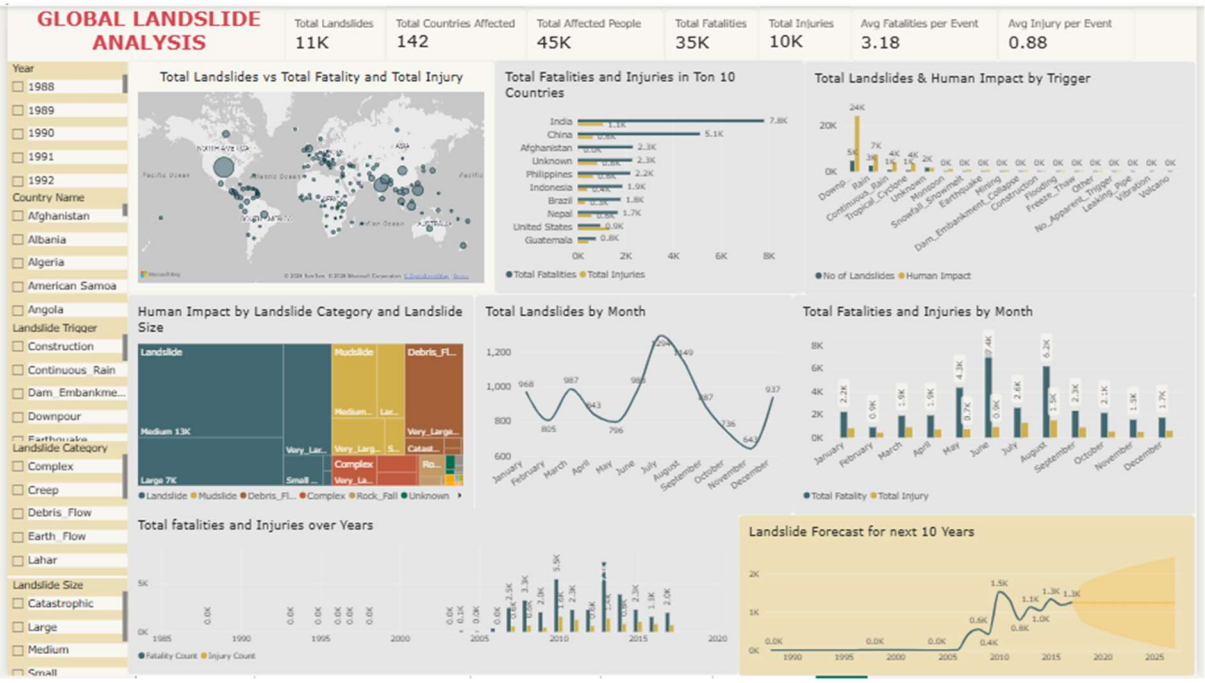
- KPI cards are displaying the values of Total landslides occurred, Total countries affected, Total affected people, Total fatalities, Total injuries, Avg fatality per event, Average injury per event over the years between 1998 to 2017
- The Geographical map is showing the values of the Total landslides happened, Total fatalities and Total injuries in all affected countries
- Clustered Bar chart is used for filtering top 10 most affected countries
- Again, Clustered Bar chart is used for analysing the details of total landslides and total human impact by landslide triggers
- Tree map chart is used for analysing Human impact by landslide category and landslide size
- Line chart is used for analysing landslide trend over months

- Line and Clustered column chart is used for analysing total fatalities and injuries happened over months
- Clustered Column chart is used to analyse total fatalities and injuries over years
- Line chart is used for forecasting the future landslide trend for the next 10 years using forecast feature

Drill-down, Filters and Slicers

- Added slicers country name, landslide trigger, category, landslide size and year

Consolidated Report



9.Insights

Key Findings

- Totally 11033 events are occurred
- 142 countries are affected
- Totally 35k fatalities and 10k injuries identified
- Avg fatalities per event is 3.18
- Avg injuries per event is 0.88
- Highest landslide occurred country is United States with 2992 counts
- Highest Fatality happened country is India with 7812 deaths
- Downpour trigger is major cause for landslide around the world
- Landslide category and medium size caused most on human impact around 13177

- July month has the highest numbers of landslides
- June has highest numbers of causalities
- Highest injury happened in august month
- Year 2013 has highest fatality count of 7251

Key Insights

- The analysis identifies countries with the highest number of recorded landslide events and highlights geographical regions frequently affected by landslides.
- Total fatalities, injuries and affected people help measure the overall human impact of landslide incidents.
- Comparing countries based on landslide occurrences and fatalities helps identify regions experiencing a greater recorded impact.
- Comparing landslide triggers with human impact helps identify which reported triggers are associated with more severe consequences.
- Historical yearly landslide trends can be used to estimate possible future event counts.
- Historical monthly patterns can help identify months that may experience higher landslide activity.
- Historical fatality and injury trends can help identify whether human impact is increasing or decreasing over time.
- Governments and disaster management authorities should prioritize geographical regions with high recorded landslide occurrences and human impact.
- Develop effective early warning systems in landslide-prone areas, particularly during historically high-risk periods.
- Regularly monitor rainfall, soil moisture, slope stability and other relevant environmental conditions to improve landslide risk assessment.
- Prioritize rescue teams, medical facilities and emergency supplies in regions with historically higher human impact

10.Conclusion

This project is helped to analysing and understanding landslide events occurred globally. And what is the main reason for the landslide, what trigger the landslide most and what are all sizes of the landslides. And also finds which months are affected mostly by landslides and how much people totally affected every year. From this we can forecast the near future landslides over the years